SECTION A (24 MARKS)
Answer All Questions

Question One

Choose a correct answer from the following questions **[@ 1 mark, Total 10 marks]**

i. Which of the following is NOT a mode of digital communication?
A. Broadcasting
B. Point-to-point communication
C. Satellite communication
D. Telecommunication

ii. Which of the following is an example of an analog information source?
A. Computer data
B. Text
C. Speech signal
D. Binary code

iii. Which process involves sampling, quantization, and encoding?
A. Source encoding
B. Channel encoding
C. Formatting
D. Modulation

iv. Asynchronous transmission is best suited for which of the following?
A. High-speed communication
B. Low-speed communication
C. Real-time audio and video
D. Large data block

v. Which error correction method involves the receiver guessing the message using redundant bits?
A. Forward error correction
B. Retransmission
C. Parity check
D. Checksum

vi. What is a key characteristic of synchronous transmission?
A. Arbitrary delay between data items
B. Continuous data flow without gaps
C. Fixed intervals between data items
D. Use of start and stop bits

For a line code, the transmission bandwidth must be
A. Maximum possible
B. As small as possible
C. Depends on the signal
D. None of the above

In polar RZ format for coding, symbol '0' is represented by
A. Zero voltage
B. Negative voltage
C. Pulse is transmitted for half the duration
D. Both b and c are correct

Which element of a digital communication system is responsible for converting the signal into a form suitable for transmission?
A. Source encoder
B. Channel encoder
C. Modulator
D. Transmitter

x. The format in which the positive half interval pulse is followed by a negative half interval pulse for transmission of '1' is
A. Polar NRZ format
B. Bipolar NRZ format
C. Manchester format
D. None of the above

Question Two

Write 'T' for a TRUE statement and 'F' for a FALSE statement **[@1 mark, Total 5 marks]**

i. A message signal is transmitted from the Earth station to a satellite via an uplink
ii. The minimum bandwidth of Manchester and differential Manchester is 2 times that of NRZ.
iii. Error detection mechanisms can also correct errors in the transmitted data
iv. Parallel transmission is generally more cost-effective than serial transmission
v. In TV Broadcasting, each TV channel has its own bandwidth of 6 MHz

Question Three

Define digital modulation and mention any three outstanding advantages over traditional analog system **[4 marks]**

Question Four

i. Brief explain the function of three basic elements of Communication Systems **[3 marks]**

ii. Show the relationship between the spectral efficiency and the energy per information bit according to Shannon channel capacity **[2 marks]**

SECTION B (36 MARKS)

Answer Only THREE Questions

Question Five

i. What is a sampling used in digital transmission? **[1 mark]**

ii. We have sampled a low-pass signal with a bandwidth of 200 KHz using 1024 levels of quantization.

   a) Calculate the bit rate of the digitized signal **[1 mark]**

   b) Calculate the PCM bandwidth of this signal **[1 mark]**

iii. The **figure 1** below shows a digital signal waveform, Compute

   a) Number of bits transmitted in 1 s **[1 mark]**

   b) Bit rate **[1 mark]**

   c) Number of levels **[1 mark]**

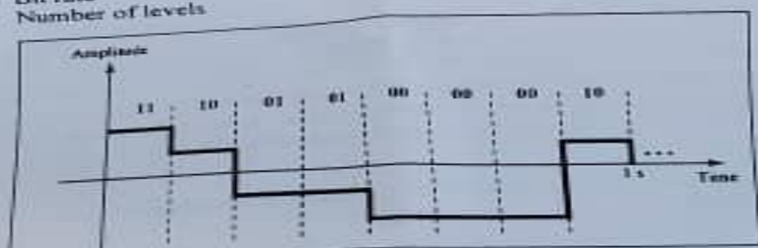

Figure 1

iv. With the clear waveform diagrams explain the three conditions of sampling process **[6 marks]**

uestion Six

i. What is a scrambling? [2 marks]

ii. Represent the bit sequence of 01001110 using both Manchester and differential Manchester encoding on the same vertical axis [4 marks]

iii. How will you encode the data stream of 01001100, using the following encoding schemes?
a) NRZ-L b) Bipolar AMI c) Pseudoternary [6 marks]

Question Seven

i. A system is using NRZ-I to transfer 1.5-Mbps data. What are the average signal rate and minimum bandwidth? [2 marks]

ii. Write the Ethernet standards of the following Signaling
a) Manchester coded b) MLT-3 c) NRZI d) PAM-5 e) NRZ [5 marks]

iii. Draw a graph of spectrum comparison of Mean square voltage per unit bandwidth versus Normalized frequency (*f/R*). *HINT: graph should include AMI, Pseudo ternary, B8ZS, HDB3, NRZ-L and NRZ-I* [5 marks]

Question Eight

i. Assume that a voice channel occupies a bandwidth of 4 KHz. We need to combine three voice channels into a link with a bandwidth of 12 KHz, from 20 to 32 KHz. Show the configuration using the frequency domain without the use of guard bands. [3 marks]

ii. Five channels, each with a 100-KHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 KHz between the channels to prevent interference? Show in the Figure the arrangement of channels and safe guards. [3 marks]

iii. The Advanced Mobile Phone System (AMPS) uses two bands. The first band, 824 to 849 MHz, is used for sending; and 869 to 894 MHz is used for receiving. Each user has a bandwidth of 30 KHz in each direction. The 3-KHz voice is modulated using FM, creating 30 KHz of modulated signal. How many people can use their cellular phones simultaneously? [2 marks]

iv. Four channels are multiplexed using TDM. If each channel sends 100 bytes/s and we multiplex 1 byte per channel, show the frame traveling on the link, the size of the frame, the duration of a frame, the frame rate, and the bit rate for the link. [4 marks]